Lesson 11.3 - Hypothesis Testing with Two Proportions

> **Video:** Hypothesis Testing With Two Proportions by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

Formulas

Quantity	Formula
Sample proportions	$\hat{p}_1 = \frac{x_1}{n_1},\ \hat{p}_2 = \frac{x_2}{n_2}$
Pooled proportion	$\bar{p} = \frac{x_1 + x_2}{n_1 + n_2}$
Test statistic	$z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\bar{p}\,(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$

Under $H_0 : p_1 = p_2$, the term $p_1 - p_2$ is **0**. The proportions are pooled because H_0 assumes both groups share one common proportion. Large samples → use the normal (z) distribution.

Example - defective laptops

Two batches of laptops were tested: 32 of 800 were defective in group 1 and 30 of 500 in group 2. Is the difference significant at $\alpha = 0.05$?

Hypotheses (the test is run because someone suspects a difference):

$$H_0 : p_1 = p_2 \qquad H_a : p_1 \neq p_2$$

Sample proportions:

$$\hat{p}_1 = \frac{32}{800} = 0.04, \qquad \hat{p}_2 = \frac{30}{500} = 0.06$$

Critical values (two-tailed, $\alpha/2 = 0.025$ per tail, left area 0.975): $z = \pm 1.96$

Pooled proportion:

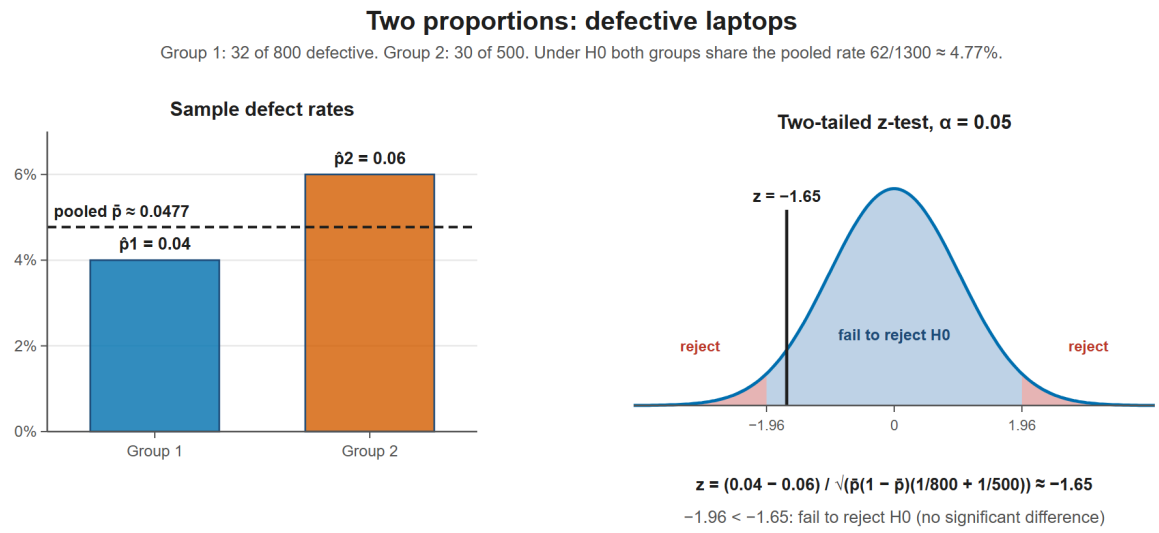
$$\bar{p} = \frac{32 + 30}{800 + 500} = \frac{62}{1300} \approx 0.04769$$

Test statistic:

$$z = \frac{0.04 - 0.06 - 0}{\sqrt{0.04769(1 - 0.04769)\left(\frac{1}{800} + \frac{1}{500}\right)}} = \frac{-0.02}{0.012149} \approx -1.65$$

$-1.96 < -1.65 < 0$, which is in the fail-to-reject region.

Fail to reject H_0. At the 5% level the difference in defect rates (4% vs 6%) is **not statistically significant**. This doesn't prove the rates are equal; the data just don't show a clear difference.



The two sample rates differ by 2 percentage points, but with these sample sizes that gives only z ≈ −1.65, which is inside ±1.96. The difference is not statistically significant.

Key takeaways

- $H_0 : p_1 = p_2$. Use the **pooled** proportion $\bar{p} = (x_1 + x_2)/(n_1 + n_2)$ in the standard error.
- $z = (\hat{p}_1 - \hat{p}_2)/\sqrt{\bar{p}(1-\bar{p})(1/n_1 + 1/n_2)}$.
- Compare with the critical value(s) or use a p-value. Fail to reject means "no significant difference found", not "proven equal".